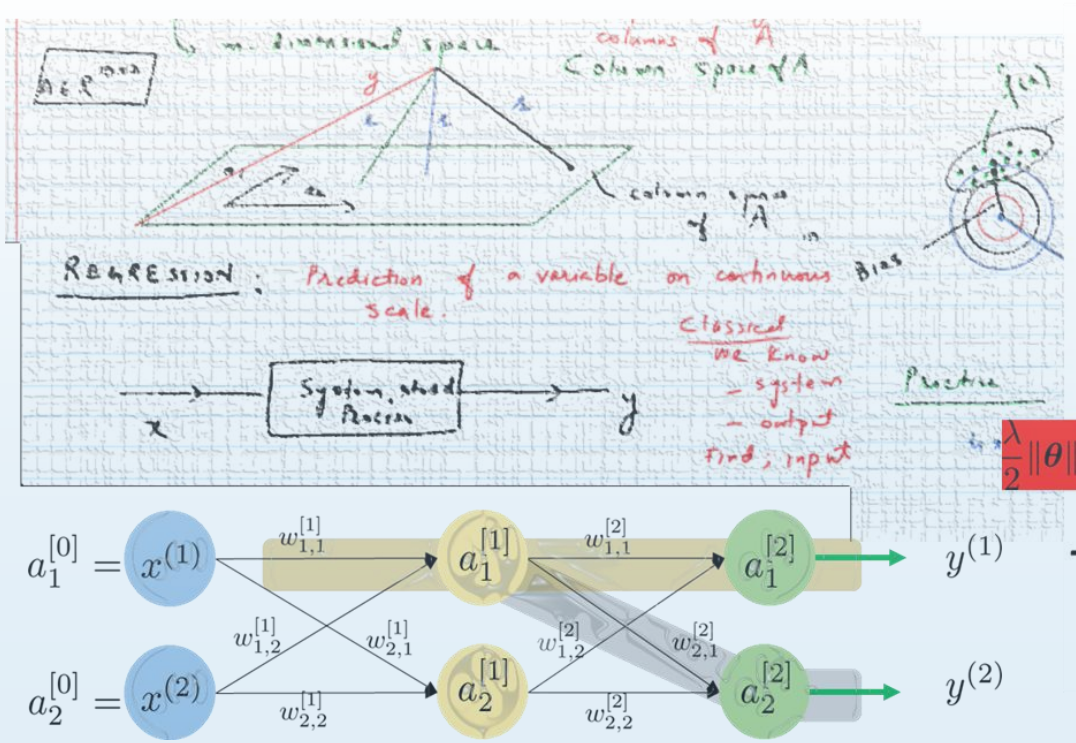
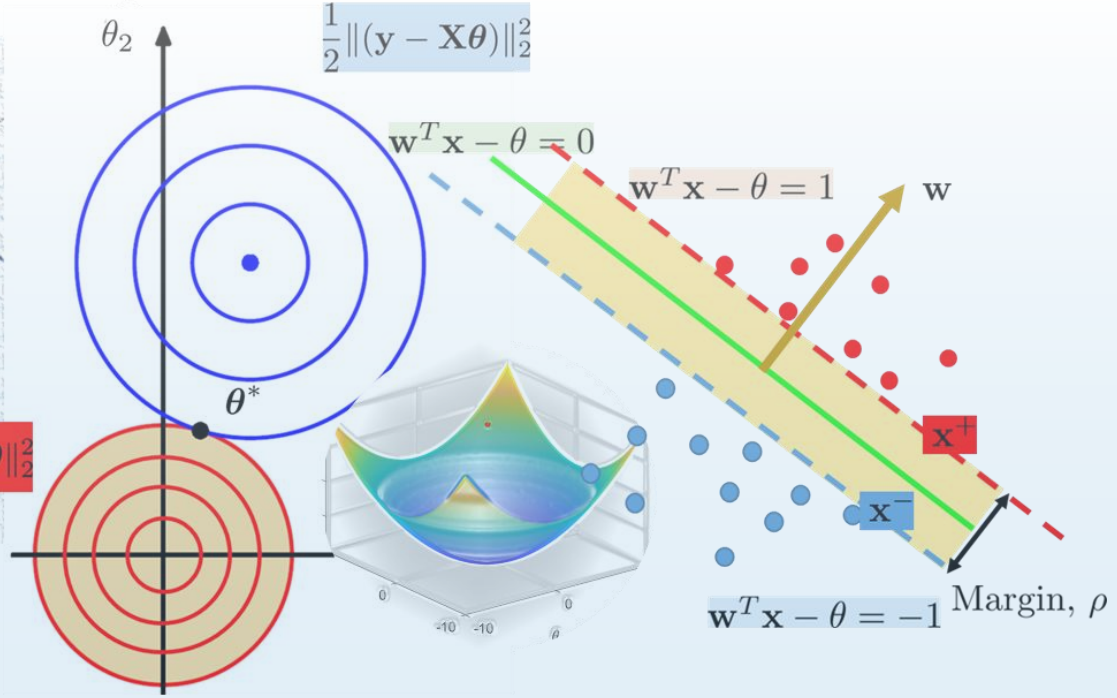




AI-501 Mathematics for AI

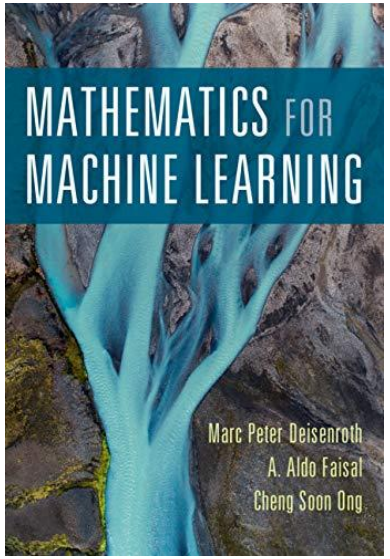
Singular Value Decomposition

Zubair Khalid
School of Science and Engineering

LUMS
A Not-for-Profit University

Outline

- Positive/negative definite and semi-definite matrices
- Singular Value Decomposition (SVD)
  - Formulation
  - Interpretation
  - Application examples



Chapter 3.2.3

Positive/Negative Definite/Semi-Definite Matrices

Definition:

For a matrix $A \in \mathbf{R}^{n \times n}$, if

$$x^T Ax \geq 0 \qquad \forall\ x \in \mathbf{R}^n \qquad A \text{ is positive semi-definite (PSD)}$$

$$x^T Ax > 0 \qquad \forall\ x \in \mathbf{R}^n \qquad A \text{ is positive definite (PD)}$$

$$x^T Ax \leq 0 \qquad \forall\ x \in \mathbf{R}^n \qquad A \text{ is negative semi-definite (NSD)}$$

$$x^T Ax < 0 \qquad \forall\ x \in \mathbf{R}^n \qquad A \text{ is negative definite (ND)}$$

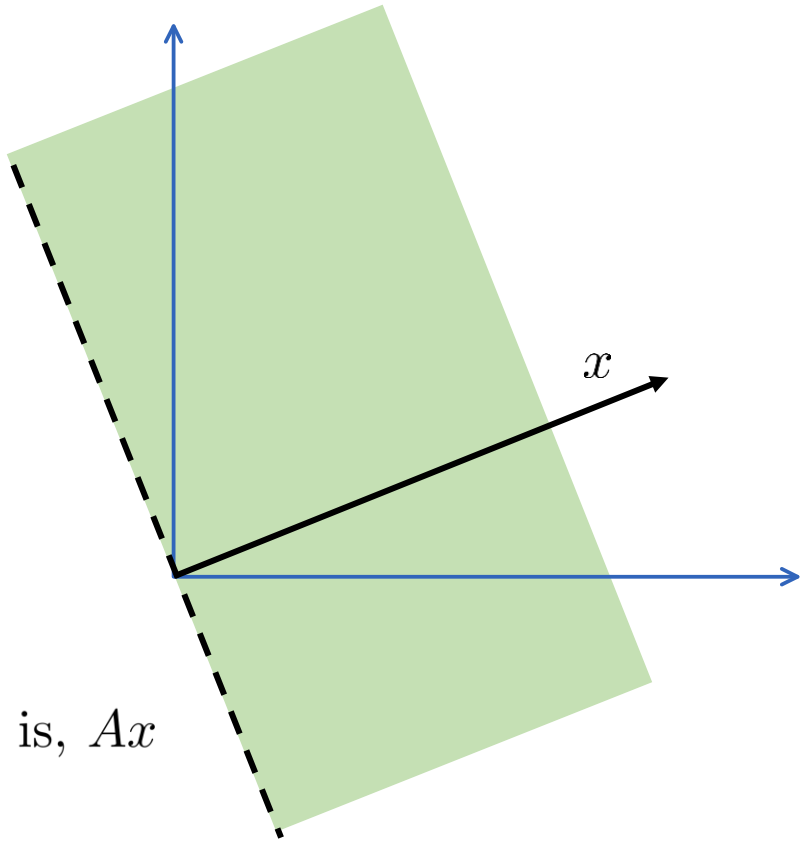
Positive Definite and Semi-Definite Matrices

Interpretation:

A is positive semi-definite (PSD)

$$x^T Ax \geq 0$$

- Let $y = Ax$
  - y is a linear transformation defined by the matrix A.
- $x^T y \geq 0$ implies angle between x and y is less than or equal to $\frac{\pi}{2}$.
- $x^T y \geq 0$ implies angle between x and linearly transformed x, that is, Ax is less than or equal to $\frac{\pi}{2}$.



Graphically, a vector x when transformed by a matrix A, that is, Ax can be anywhere in the green region including the dashed boundary where $x^T Ax = 0$

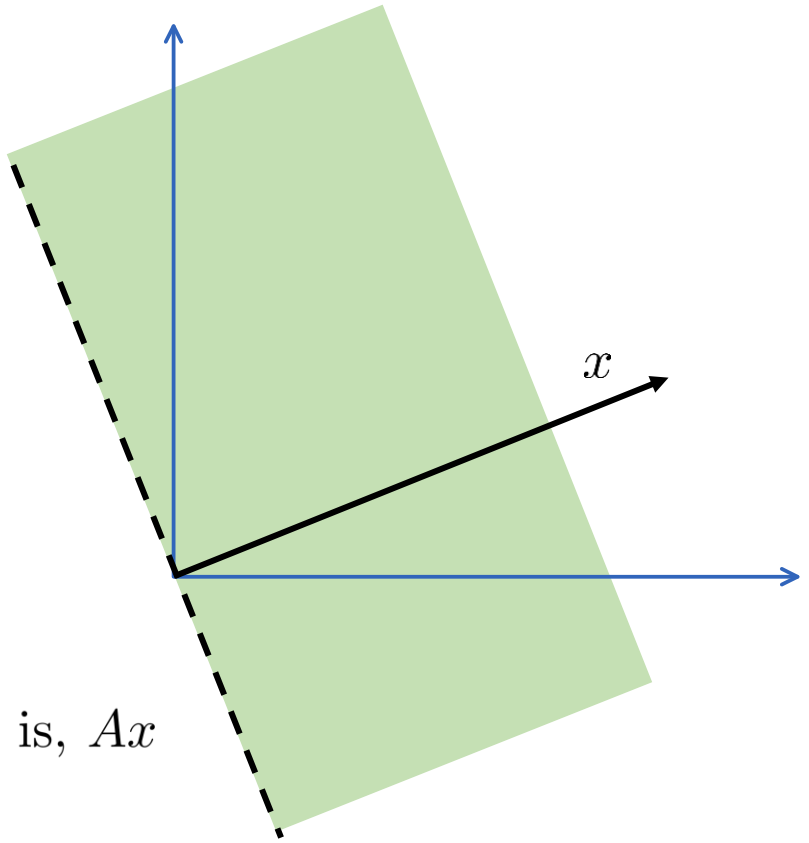
Positive Definite and Semi-Definite Matrices

Interpretation:

A is positive definite (PD)

$$x^T Ax > 0$$

- Let $y = Ax$
  - y is a linear transformation defined by the matrix A.
- $x^T y > 0$ implies angle between x and y is less than $\frac{\pi}{2}$.
- $x^T y \geq 0$ implies angle between x and linearly transformed x, that is, Ax is less than $\frac{\pi}{2}$.



Graphically, a vector x when transformed by a matrix A, that is, Ax can be anywhere in the green region excluding the dashed boundary where $x^T Ax = 0$

Positive Definite and Semi-Definite Matrices

Eigenvalues of symmetric PSD/PD matrix:

For a symmetric and PD matrix A, eigenvalues are positive.

How?

- We already know that the eigenvalues of a symmetric matrix are real.
- For a PD symmetric, we require $x^T Ax > 0$
- If we take $x = q$, where q is an eigenvector with an associated eigenvalue λ

$$q^T Aq > 0 \Rightarrow \lambda q^T q > 0 \Rightarrow \lambda \|q\|_2^2 > 0 \Rightarrow \lambda > 0$$

Similarly, we can show the following:

For a symmetric and PSD matrix A, eigenvalues are non-negative.

For a symmetric and NSD matrix A, eigenvalues are non-positive.

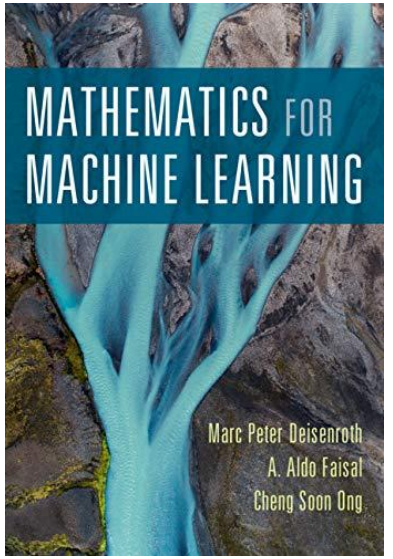
For a symmetric and ND matrix A, eigenvalues are negative.

Outline

- Positive/negative definite and semi-definite matrices
- Singular Value Decomposition (SVD)

Outline

- Positive/negative definite and semi-definite matrices
- Singular Value Decomposition (SVD)
  - Formulation
  - Interpretation
  - Application examples
- Column space and Null Space



Chapter 4

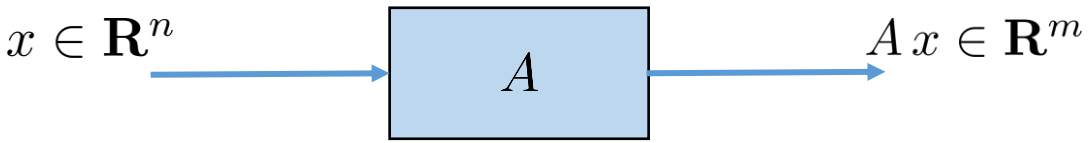
Singular Value Decomposition

Overview:

- The singular value decomposition (SVD) of a matrix is a central matrix decomposition method in linear algebra.

- It has been referred to as the "**fundamental theorem of linear algebra**" (Strang, 1993) because it can be applied to all matrices, not only to square matrices, and it always exists.

- For $A \in \mathbf{R}^{m \times n}$, we have



- SVD explains the underlying geometry of this linear transformation.

Singular Value Decomposition

Formulation:

- For any matrix $A \in \mathbf{R}^{m \times n}$, we have a singular value decomposition (SVD) given by

$$A = U \Sigma V^T$$

- Matrix $U \in \mathbf{R}^{m \times m}$ is an orthonormal matrix.
- Matrix $V \in \mathbf{R}^{n \times n}$ is an orthonormal matrix.
- Matrix $\Sigma \in \mathbf{R}^{m \times n}$ is a (special) diagonal matrix.

$m < n$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 & \dots & 0 \\ 0 & 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix}$$

$m = n$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n \end{bmatrix}$$

$m > n$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

Singular Value Decomposition

Formulation:

$$A = U \Sigma V^T$$

- Columns of U are referred to as left singular vectors of matrix A.
- Columns of V are referred to as right singular vectors of matrix A.
- $\sigma_1, \sigma_2, \ldots, \sigma_{\min(m,n)}$ are singular values of matrix A, which are (usually) indexed such that

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{\min(m,n)} \geq 0$$

Singular Value Decomposition

How to Compute SVD?

- For a matrix $A \in \mathbf{R}^{m \times n}$, we define a matrix $G = AA^T$.
- Using $A = U \Sigma V^T$, we can write G as

$$G = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T$$

- What is special about matrix G?
  - G is symmetric by definition.
  - G is positive semi-definite. **How?** You are fully equipped to show this.
- We note that $\Sigma\Sigma^T$ is a diagonal matrix of size $m \times m$.
- Eigenvalue decomposition of G gives columns of U as eigenvectors and diagonal entries of $\Sigma\Sigma^T$ as eigenvalues.
- In other words, left singular vectors of A are eigenvectors of AA^T and $\sigma^2 = \lambda$ (eigenvalue of AA^T). Furthermore, $\lambda \geq 0$ since $G = AA^T$ is PSD.

Eigenvalue decomposition of AA^T gives m left singular vectors of A and first m singular values.

Singular Value Decomposition

How to Compute SVD?

- Now we define a matrix $G = A^T A$.
- Using $A = U \Sigma V^T$, we can write G as

$$G = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$$

- What is special about matrix G?
  - G is symmetric by definition.
  - G is positive semi-definite. ***How?*** *You are fully equipped to show this.*
- We note that $\Sigma^T \Sigma$ is a diagonal matrix of size $n \times n$.
- Eigenvalue decomposition of G gives columns of V as eigenvectors and diagonal entries of $\Sigma^T \Sigma$ as eigenvalues.
- In other words, right singular vectors of A are eigenvectors of $A^T A$ and $\sigma^2 = \lambda$ (eigenvalue of $A^T A$). Furthermore, $\lambda \geq 0$ since G is PSD.

Eigenvalue decomposition of $A^T A$ gives n right singular vectors of A and first n singular values.

Now you can explain the non-negativity of the singular values.

Singular Value Decomposition

SVD Summary

- Singular value decomposition (SVD) of a matrix $A \in \mathbf{R}^{m \times n}$ is given by

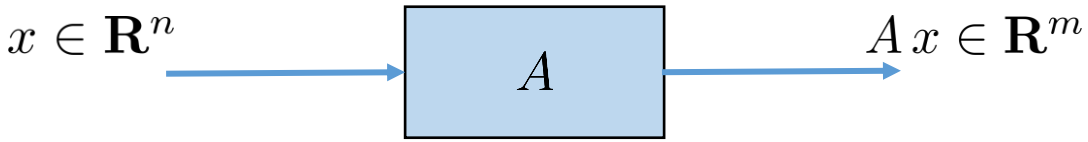
$$A = U \Sigma V^T$$

- EVD of AA^T gives U and first m singular values.
- EVD of $A^T A$ gives V and first n singular values.
- U and V are always orthogonal.
- SVD always exists.
- Singular values are non-negative, that is,

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{\min(m,n)} \geq 0$$

Singular Value Decomposition

Geometric Interpretation





Transformation (Change of Basis)

Change of basis in $\mathbf{R}^n$

Scaling (In new Basis)

Transformation (Change of Basis)

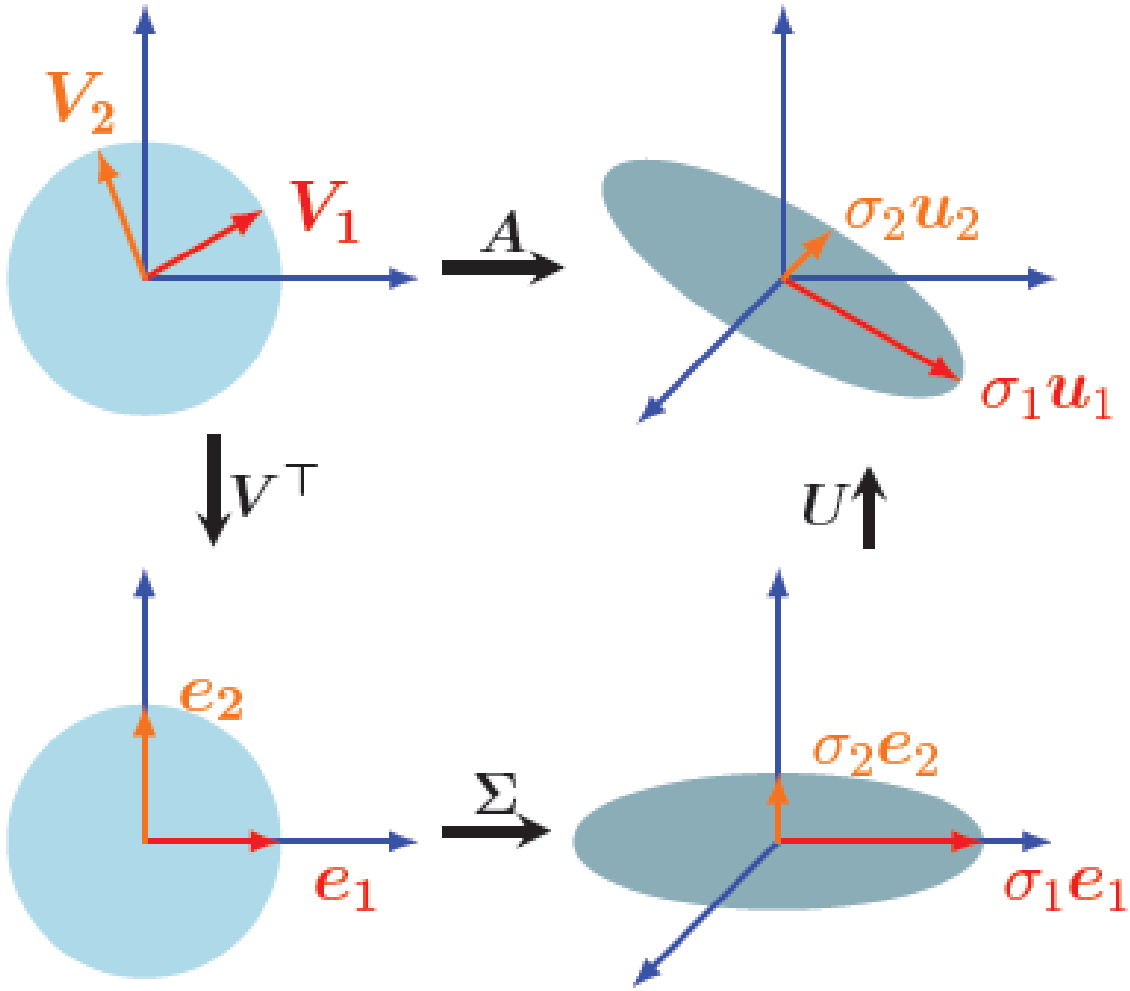
Change of basis in $\mathbf{R}^m$

Scaling along the new basis by singular values.

- $m < n$ - Drop the last $n - m$ basis (impact of columns of zeros in Σ)
- $m > n$ - Append $m - n$ basis (impact of rows of zeros in Σ)

Singular Value Decomposition

Geometric Interpretation

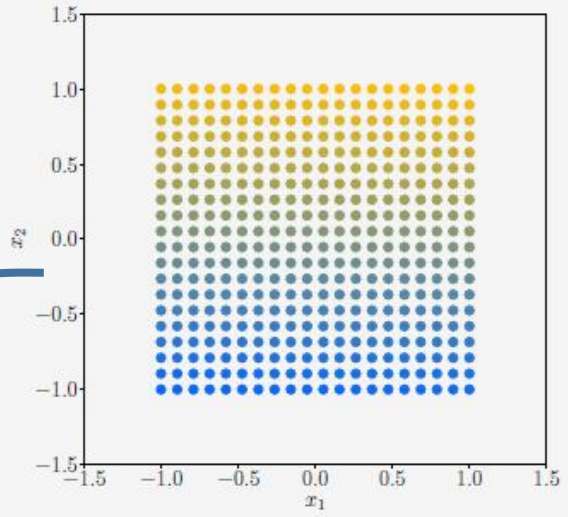


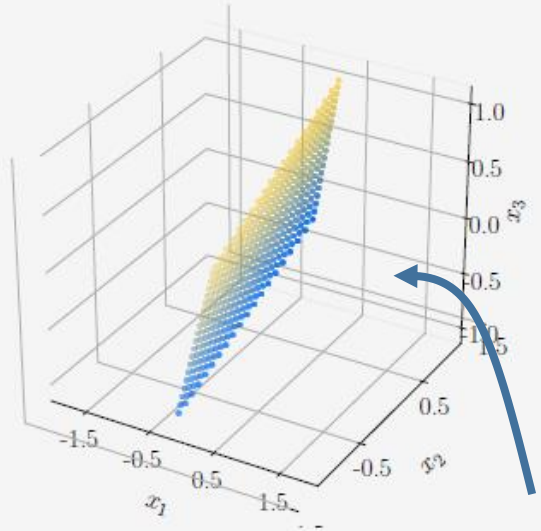
Singular Value Decomposition

Geometric Interpretation - Example

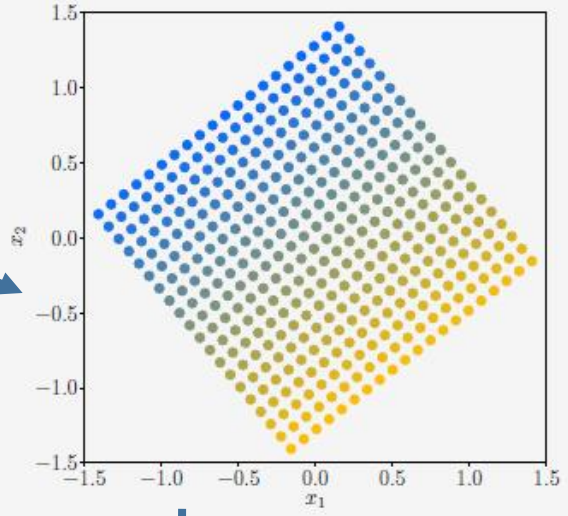
$$\boldsymbol{A} = \begin{bmatrix} 1 & -0.8 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$$

$$= \begin{bmatrix} -0.79 & 0 & -0.62 \\ 0.38 & -0.78 & -0.49 \\ -0.48 & -0.62 & 0.62 \end{bmatrix} \begin{bmatrix} 1.62 & 0 \\ 0 & 1.0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -0.78 & 0.62 \\ -0.62 & -0.78 \end{bmatrix}$$

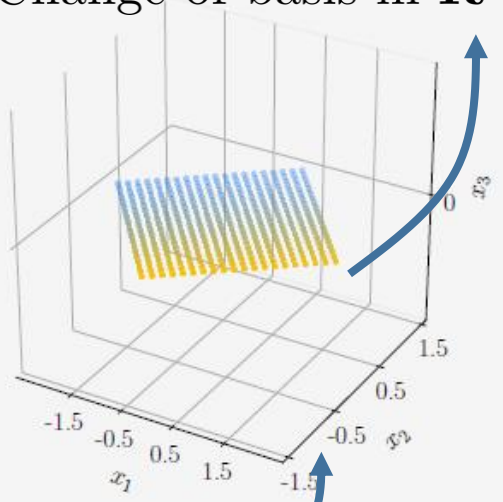




Change of basis in $\mathbf{R}^2$



Change of basis in $\mathbf{R}^3$



- Append 1 more basis
  (impact of a row of zeros in Σ)

Singular Value Decomposition

Rank of a Matrix:

- The rank of a matrix is equal to the number of non-zero singular values.

How?

- Since $A^T A$ and A have the same rank and we know that the rank of any square matrix equals the number of nonzero eigenvalues.

Application Example – Rank Estimation:

We use SVD for the estimation of rank while analyzing data. ***How?***

- Suppose that we have n data points $a_1, a_2, \ldots, a_n$, all of which live in $\mathbf{R}^m$, where n is much larger than m. Let A be the $m \times n$ matrix with columns $a_1, a_2, \ldots, a_n$.
- Assume that the the data points satisfy some linear relations, such that $a_1, a_2, \ldots, a_n$ all lie in an r dimensional subspace of $\mathbf{R}^m$. Then we would expect the matrix A to have rank r.
- If the data points are obtained from measurements with errors, then the matrix A will probably have full rank m. But only r of the singular values of A will be large, and the other singular values will be close to zero.
- Using SVD, we can can estimate an "approximate rank" of A by counting the number of singular values which are much larger than the others.

Singular Value Decomposition

Application: Matrix Approximation

- A matrix $A \in \mathbf{R}^{m \times n}$ can be decomposed using SVD as

$$A = U \Sigma V^T = \sum_{i=1}^{\min(m,n)} u_i \sigma_i v_i^T$$

- If rank of a matrix is $r \leq \min(m, n)$, we can truncate the summationion at r

$$A = \sum_{i=1}^{r} u_i \sigma_i v_i^T$$

- Using SVD formulation, we can define k rank approximation of the matrix A by including first k singular vectors and associated singular values in the representation, that is,

$$A \approx \sum_{i=1}^{k} u_i \sigma_i v_i^T$$

(k-rank approximation)

Singular Value Decomposition

- Demo